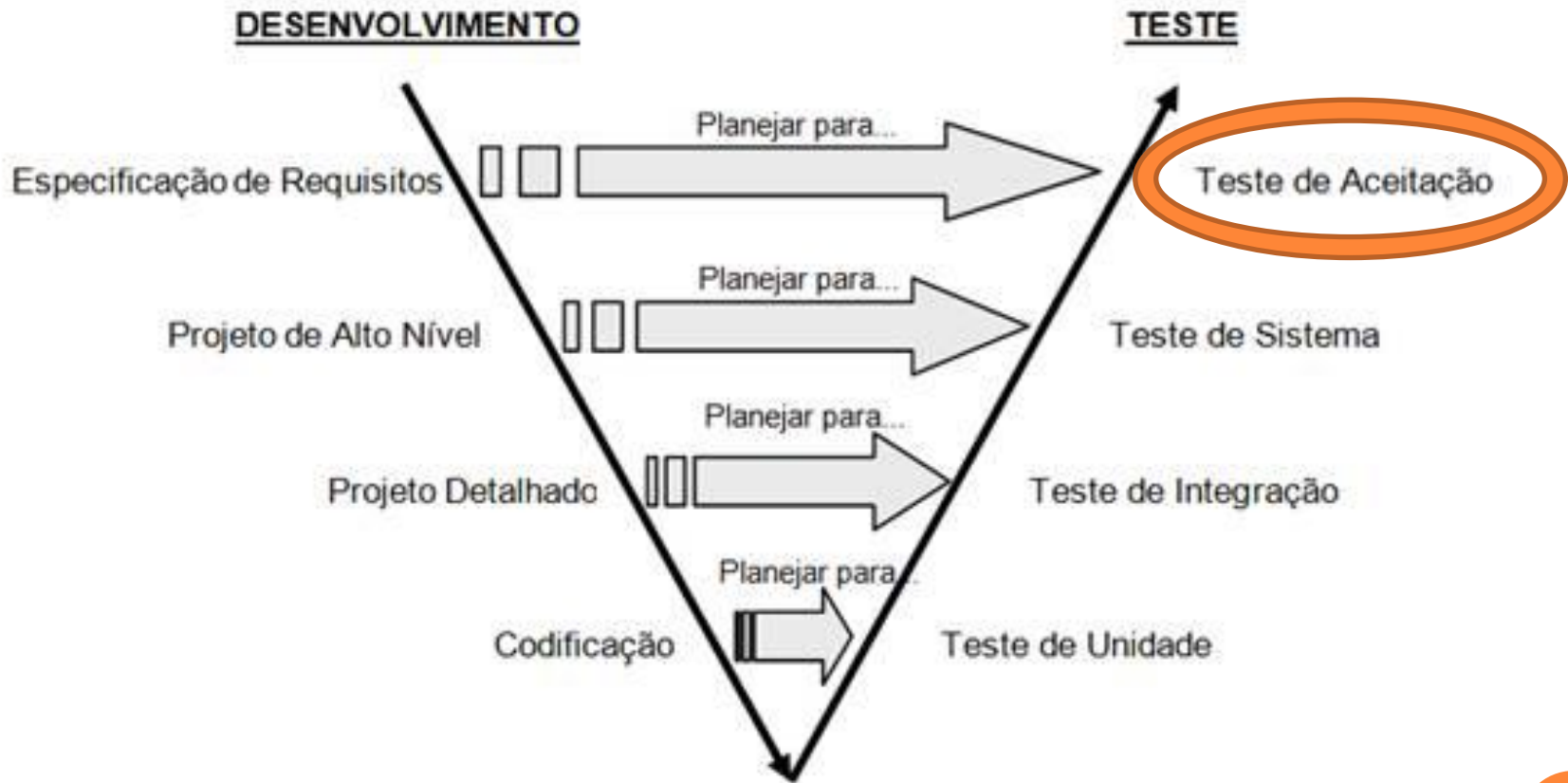


BDD

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TESTE EM V



TESTE DE ACEITAÇÃO

- Definem os testes que devem passar para determinar que o sistema está pronto
- São baseados em critérios de aceitação
- Tem um nível de abstração maior e são definidos com cliente, pessoas de negócio, desenvolvedores e testadores.



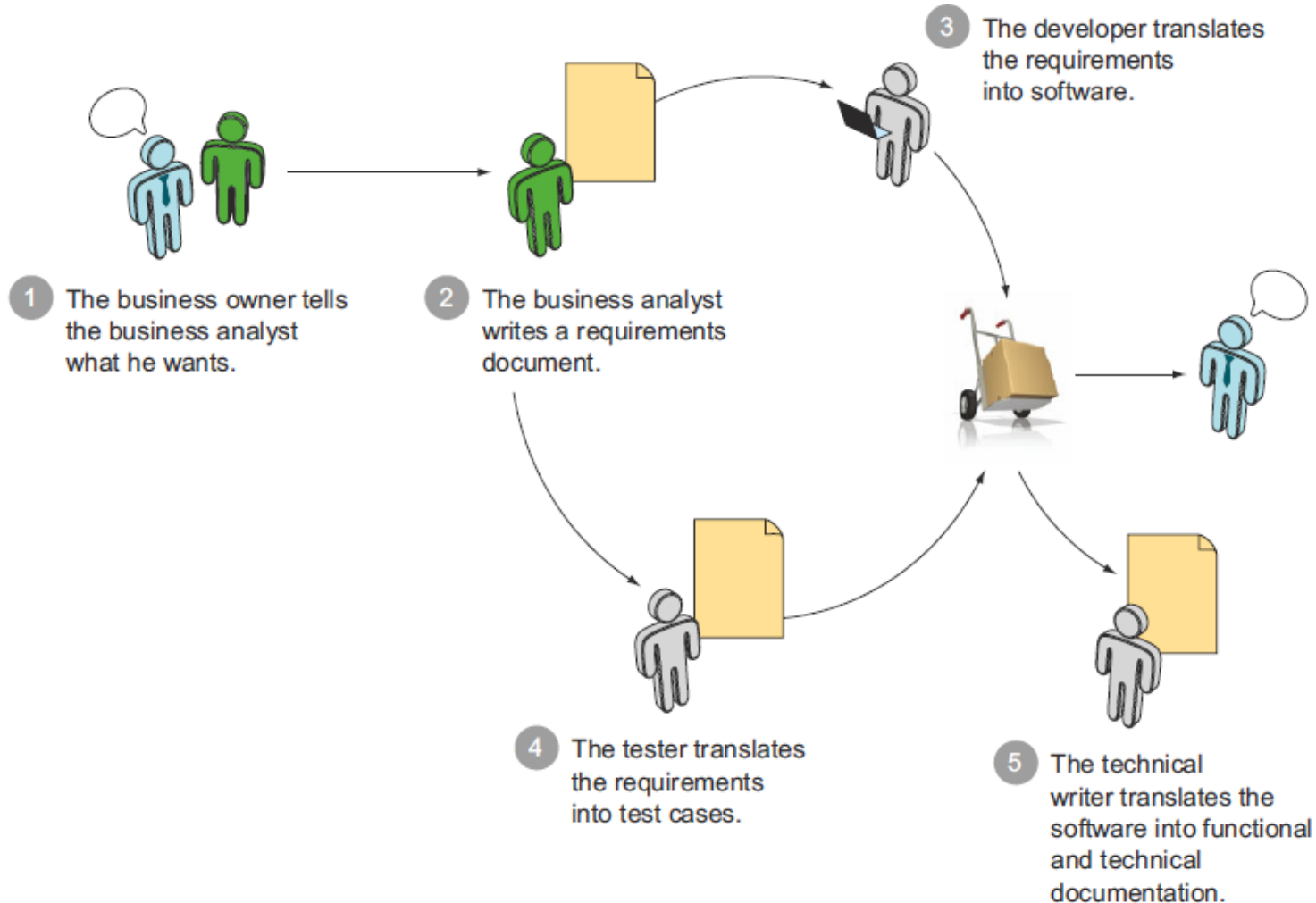


Figure 1.1 The traditional development process provides many opportunities for misunderstandings and miscommunication.



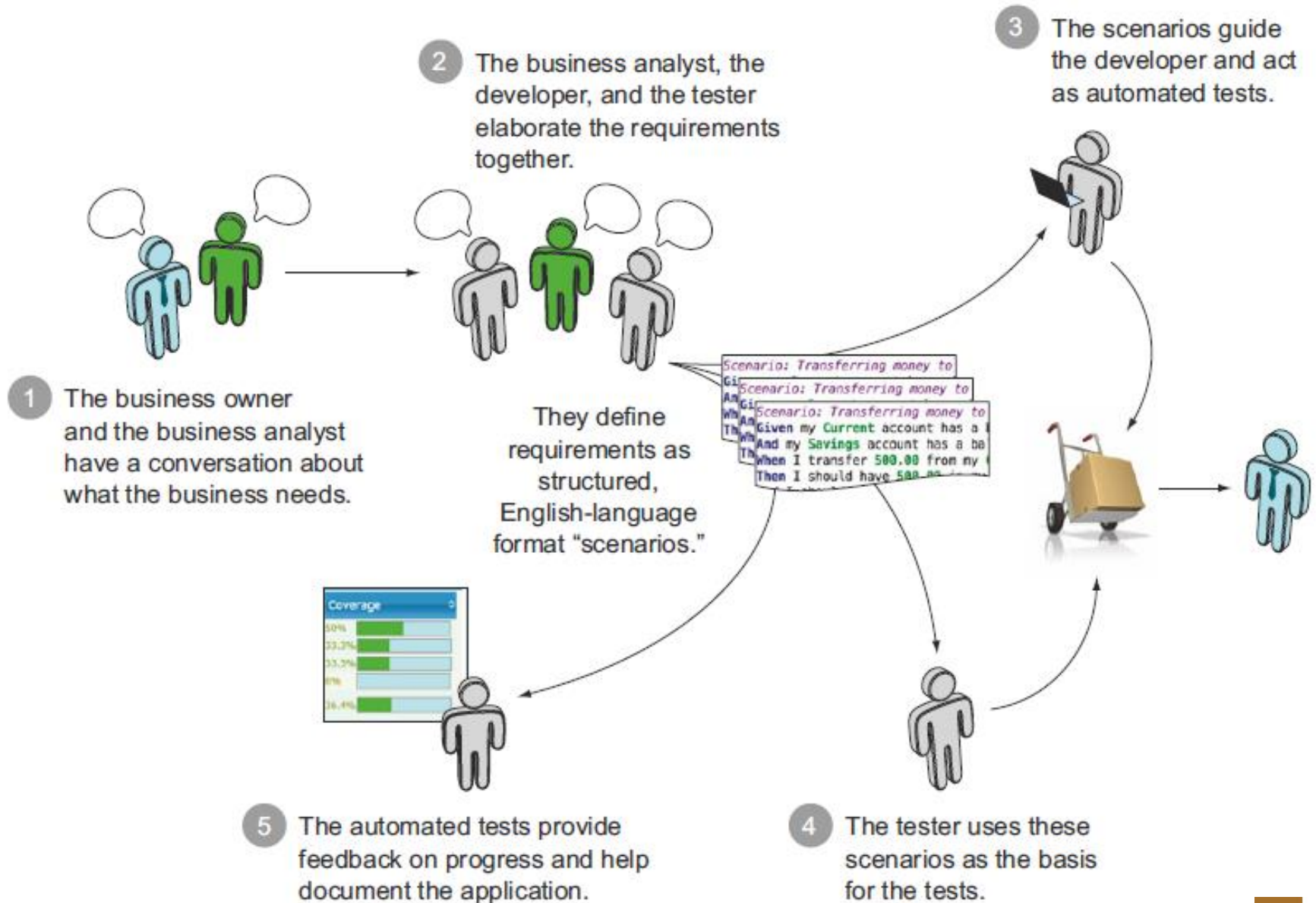
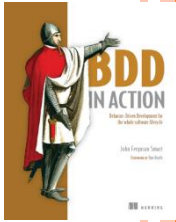


Figure 1.2 BDD uses conversations around examples, expressed in a form that can be easily automated, to reduce lost information and misunderstandings.



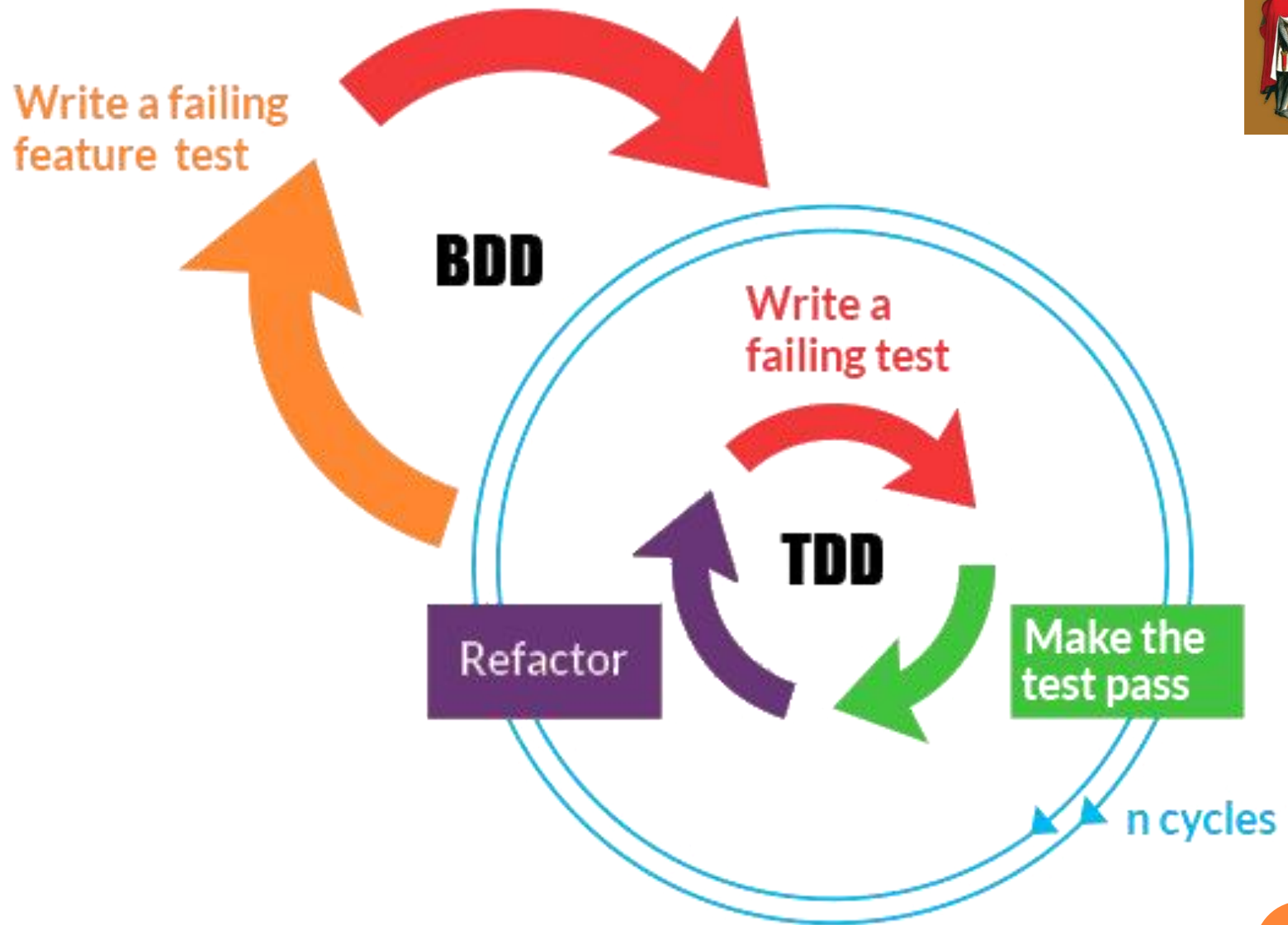
HISTÓRIA DE USUÁRIO VIRAM TESTES



BDD

- Behavior-Driven Development (BDD) é um conjunto de práticas de engenharia de software que tem o objetivo de ajudar equipes a construir e entregar software de grande valor e qualidade mais rápido.
- Ela se apoia em outras práticas como o TDD, tendo sido proposta inicialmente como um complemento à essa prática.
- Ela provê uma linguagem comum baseada em sentenças estruturadas em linguagem de alto nível que facilitam a comunicação entre membros da equipe de desenvolvimento e stakeholders de negócio.





DISCUTINDO HISTÓRIAS

Users are required to use secure passwords (strings of at least 6 characters with at least one letter, one number, and one symbol)

“What should happen if a user enters an insecure password?”

“Can you give me examples of passwords you consider secure and insecure?”

“What are examples of ‘symbols’?”

“What about spaces?”

“What about dictionary words with obvious substitutions that meet the criteria but still may be insecure, like ‘p@ssw0rd’?”

“What about existing accounts?”

“How will you know this feature ‘works’?”

A decorative graphic on the left side of the slide. It features a series of vertical stripes in shades of brown, tan, and grey. Overlaid on these stripes are several orange circles of varying sizes, arranged in a cluster that tapers towards the bottom.

PENSAR EM CENÁRIOS

→ História de usuário

Feature: Transferring money between accounts

In order to manage my money more efficiently

As a bank client

I want to transfer funds between my accounts whenever I need to

Scenario: Transferring money to a savings account

Given my Current account has a balance of 1000.00

And my Savings account has a balance of 2000.00

When I transfer 500.00 from my Current account to my Savings account

Then I should have 500.00 in my Current account

And I should have 2500.00 in my Savings account

Scenario: Transferring with insufficient funds

Given my Current account has a balance of 1000.00

And my Savings account has a balance of 2000.00

When I transfer 1500.00 from my Current account to my Savings account

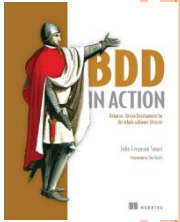
Then I should receive an 'insufficient funds' error

Then I should have 1000.00 in my Current account

And I should have 2000.00 in my Savings account

→ Cenários no formato BDD





FORMATO BDD

- Given: define as precondições do cenário e prepara o teste
- When: descreve as ações que serão executadas no teste
- Then: descreve o resultado esperado para esse cenário



FORMATO BDD

- A descrição pode ser genérica suportando diferentes valores de entrada

Scenario Outline: Earning interest

Given I have an account of type <account-type> with a balance of <initial-balance>

When the monthly interest is calculated

Then I should have earned at an annual interest rate of <interest-rate>

And I should have a new balance of <new-balance>

Examples:

initial-balance	account-type	interest-rate	new-balance
10000	current	1	10008.33
10000	savings	3	10025
10000	supersaver	5	10041.67

- Nesse caso, seriam executados 3 testes, um para cada conjunto de entradas- resultados esperados





**DEPOIS QUE OS CENÁRIOS
ESTIVEREM DEFINIDOS EM ALTO
NÍVEL, ELES PRECISAM SER
IMPLEMENTADOS**

EXEMPLO COM JBEHAVE

```
@Given("my $accountType account has a balance of $amount")
public void setupInitialAccount(AccountType accountType, double amount) {
    Account account = Account.ofType(accountType).withInitialBalance(amount);
    accountService.create(account);
    myAccounts.put(accountType, account.getAccountNumber());
}

@When("I transfer $amount from my $source account to my $destination account")
public void transferAmountBetweenAccounts(double amount,
                                           AccountType source,
                                           AccountType destination) {
    Account sourceAccount = accountService.findByNumber(myAccounts.get(source)).getAccount();
    Account destinationAccount = accountService.findByNumber(myAccounts.get(destination)).getAccount();
    accountService.transfer(amount).from(sourceAccount).to(destinationAccount);
}
```





MAS CALMA. O MAIS
IMPORTANTE É ENTENDER A
DEFINIÇÃO DE CENÁRIOS

Feature: Earning extra points from Frequent Flyer status

Scenario: A standard Frequent Flyer earns the base point value of a trip

Given I am a Standard Frequent Flyer member

When I fly on a flight that is worth 439 base points

Then I should earn a total of 439 points

Scenario: A Silver Frequent Flyer will earn a 50% status bonus

Given I am a Silver Frequent Flyer member

When I fly on a flight that is worth 439 base points

Then I should earn a status bonus of 220 points

And I should earn a total of 659 points

Scenario: A Gold Frequent Flyer will earn a 75% status bonus

Given I am a Gold Frequent Flyer member

When I fly on a flight that is worth 2040 base points

Then I should earn a status bonus of 1530 points

And I should earn a total of 3570 points

Scenario: A Gold Frequent Flyer benefits from a guaranteed minimum of 1000 points per trip

Given I am a Gold Frequent Flyer member

When I fly on a flight that is worth 439 base points

Then I should have a guaranteed minimum of 1000 earned points per trip

And I should earn a total of 1000 points

BDD EXEMPLO

- **As a Scrum Master I want to see Lead/Cycle time progress**
- As a Scrum Master
I want to see Lead/Cycle time progress
So that I know whether we are improving our development process or not
- **Scenario #1**
Given Reports section in project and Bug Tracking practice is disabled
When I navigate to Lead and Cycle Time Report
Then I see Lead Time chart
And chart contains 1 line for stories
- **Scenario #2**
Given Reports section in project and Bug Tracking practice is disabled
When I navigate to Lead and Cycle Time Report
Then I see Cycle Time
And chart contains 1 line for stories
- **Scenario #3**
Given Reports section in project and Bug Tracking practice is ENABLED
When I navigate to Lead and Cycle Time Report
Then I see Lead Time chart
And chart contains 2 lines (for stories and bugs)
- **Scenario #4**
Given Reports section in project and Bug Tracking practice is ENABLED
When I navigate to Lead and Cycle Time Report
Then I see Cycle Time
And chart contains 2 lines (for stories and bugs)



EXERCÍCIO

- Pensar em cenário para a seguinte história de usuário
- Para visitar minha família
- Como viajante
- Desejo saber qual o próximo ônibus saindo de Quixadá para Fortaleza

